Priyanshu Singh

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SUMMARY

Aspiring Software Developer with strong expertise in Data Structures & Algorithms and backend development using C++ and Python. Experienced in building scalable graph-based systems and data-driven applications. Seeking a Software Development Internship to contribute to high-impact engineering projects.

EDUCATION

B.Tech in Computer Science Engineering, Jaypee Institute of Information Technology (2024–2028)

TECHNICAL SKILLS

Languages: C++, Python, SQL, JavaScript

Libraries/Tools: Pandas, NumPy, Matplotlib, NetworkX, Scikit-learn

Concepts: Data Structures & Algorithms, Graph Algorithms, Backend Development

PROJECTS

Smart City Infrastructure Analyzer

• Engineered a graph-based infrastructure system handling 10,000+ nodes and edges for urban network simulation.

• Implemented Dijkstra, BFS, MST, and Tarjan's Algorithm, improving path computation efficiency by ~35%.

• Designed dynamic update handling for real-time road changes, reducing recalculation overhead by ~40%.

• Integrated Python visualization (NetworkX, Matplotlib) for real-time graph rendering and traffic analysis.

AQI Data Analytics Dashboard

• Processed and analyzed 50,000+ data records across multiple cities for pollution trend analysis.

• Performed data cleaning, normalization, and standardization improving data quality by ~30%.

• Built visual dashboards (heatmaps, trends) enabling identification of key pollutants and seasonal patterns.

• Reduced analysis time by ~25% using optimized Pandas operations.

Software Metrics & Effort Estimation Platform

• Developed a web-based estimation tool implementing LOC, Function Point Analysis, and COCOMO models.

• Automated effort and cost estimation reducing manual calculation time by ~50%.

• Designed interactive UI improving usability and input efficiency.

ACHIEVEMENTS (ADD YOURS)

- Solved 200+ DSA problems on LeetCode (update)
- Active problem solver with focus on graph algorithms